No Andronov–Hopf bifurcation, no periodic orbit: cases 9 and 10 of Banaji–Boros–Hofbauer

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Solver: Claude Opus 5.5 (effort xhigh), 24 Oct 2026 · Independent verification: Claude Opus 5.5 (effort xhigh), fresh context, 25 October 2026 (confirmed with corrections)

Abstract. Banaji, Boros and Hofbauer classified the two-species, three-reaction mass-action networks with at most bimolecular sources that admit an Andronov–Hopf bifurcation. Apart from the vertical (degenerate) bifurcation of their case 8, such networks occur only in cases 9 and 10, with source complexes $2X, X+Y, Y$ and $2X, X+Y, 0$ and arbitrary targets. They asked whether the networks of these cases that admit no Andronov–Hopf bifurcation can nevertheless have a limit cycle. We show that they cannot: for every such network and every choice of rate constants, the mass-action system has no periodic orbit in the closed positive quadrant. Hence, granting their supercriticality computation, a network of case 9 or 10 has a limit cycle for some rate constants if and only if it admits an Andronov–Hopf bifurcation. The proof first reduces everything to three explicit families. These are linear in y, and become Liénard systems after a shear along the x-nullcline. A comparison lemma for Liénard systems then reduces non-existence to a single inequality that does not involve the rate constants, which we prove in closed form.

The question

Banaji, Boros and Hofbauer [1] (BBH) list the ten configurations of three non-collinear, at most bimolecular source complexes on two species. Their Theorem 9 (arXiv numbering; Theorem 3 of the published version, with identical text) decides, for each configuration, which networks admit an Andronov–Hopf bifurcation. Apart from case 8, where the bifurcation is vertical (a centre at the critical parameter), only cases 9 ($2X, X+Y, Y$) and 10 ($2X, X+Y, 0$) contain such networks, and there the bifurcation is supercritical. Lemma and theorem numbers of [1] refer to arXiv v1 throughout; in the published version arXiv Lemmas 2, 3, 4 are Lemmas 1, 2, 3 and Theorem 9 is Theorem 3. In their Conclusions (§6) they write:

> *"However, we leave it open whether the networks in cases 9 and 10 admitting no Andronov–Hopf bifurcations can have a limit cycle. For example, we do not know whether the system $2X \to 3X+2Y$, $X+Y \to 0$, $Y \to X+Y$, $\dot{x} = \kappa_1 x^2 - \kappa_2 xy + \kappa_3 y$, $\dot{y} = 2\kappa_1 x^2 - \kappa_2 xy$ admits a limit cycle."*

The example was settled in [2] with a Dulac function. Here we settle the general question.

Theorem 1. Let $\mathcal{N}$ be a three-reaction mass-action network on two species whose sources are $2X, X+Y, Y$ (case 9) or $2X, X+Y, 0$ (case 10), with arbitrary target complexes. If $\mathcal{N}$ does not satisfy the Andronov–Hopf condition of Theorem 9 [1], then for every $\kappa \in \mathbb{R}^3_{>0}$ its mass-action system has no periodic orbit in $\mathbb{R}^2_{\geq 0}$. In particular it has no limit cycle. The same holds after exchanging X and Y (BBH

list the cases up to this exchange).

Corollary 2. For a network of case 9 or 10, consider: (a) a periodic orbit exists for some κ; (b) a limit cycle exists for some κ; (c) BBH's Andronov–Hopf condition holds. Then (a)⇒(c) by Theorem 1, and (b)⇒(a) trivially. BBH state, from a computation of the first focal value, that the bifurcation is supercritical, which gives (c)⇒(b). Granting that computation, which we did not re-check, the three statements are equivalent.

1 Setting

Following (13)–(18) [1], reaction i is $a_iX + b_iY \to (a_i + c_i)X + (b_i + d_i)Y$, with integer $(c_i, d_i) \neq 0$ and nonnegative target; Γ is the 2×3 stoichiometric matrix with rows $c = (c_1, c_2, c_3)$ and $d = (d_1, d_2, d_3)$. The rate laws are $\kappa_i m_i$, where $m = (x^2, xy, y)$ in case 9 and $m = (x^2, xy, 1)$ in case 10. The admissible ranges are $c_1 \geq -2$, $d_1 \geq 0$, $c_2, d_2 \geq -1$, $c_3 \geq 0$, and $d_3 \geq -1$ (case 9) or $d_3 \geq 0$ (case 10). Let $u = c \times d$ and $A := c_1u_1 - c_3u_3$. This A is a number, not to be confused with the function $A(x)$ of §3.

BBH's condition is the following. In case 9: either $c_1 > 0$, $c_2 = -1$, $c_3 > 0$, $d_1 > 0$, $d_2 \geq -1$, $d_3 \geq -1$ and $\frac{1}{2}(d_3/c_3 + d_1/c_1) < d_2/c_2 < d_1/c_1$; or $c_1 > 0$, $c_2 = -1$, $c_3 = 0$, $d_1 > 0$, $d_2 \geq -1$, $d_3 = -1$ and $d_2/c_2 < d_1/c_1$. In case 10: $c_1 > 0$, $c_2 = -1$, $c_3 > 0$, $d_1 > 0$, $d_2 = -1$, $d_3 \geq 0$ and $\frac{1}{2}(d_3/c_3 + d_1/c_1) < d_2/c_2 < d_1/c_1$.

We use the following facts.

(F1) $\mathbb{R}^2_{>0}$ is forward invariant: a reaction with $c_i < 0$ has $a_i \geq 1$, and similarly in y. The boundary of the quadrant contains no Jordan curve. Hence every periodic orbit in $\mathbb{R}^2_{\geq 0}$ lies in $\mathbb{R}^2_{>0}$.

(F2) If $\operatorname{rank}\Gamma = 2$, a positive equilibrium exists only if $u \in \mathbb{R}^3_{>0} \cup \mathbb{R}^3_{<0}$, and then it is unique: $\kappa_i m_i(\bar{x}, \bar{y}) = \mu u_i$.

(F3) Put $x = \bar{x}X$, $y = \bar{y}Y$ and $t = (\bar{x}/\mu)T$. The system becomes $X' = \sum_i c_i u_i m_i(X, Y)$, $Y' = \rho \sum_i d_i u_i m_i(X, Y)$, with $\rho = \bar{x}/\bar{y} \in (0, \infty)$ and equilibrium $(1, 1)$. There, $\det J = \rho u_1 u_2 u_3$ (case 9) or $2\rho u_1 u_2 u_3$ (case 10), and $\operatorname{tr} J = A - \rho d_1 u_1$ (case 9) or $A + \rho d_2 u_2$ (case 10).

(F4) We use Poincaré index theory, the Jordan curve theorem and the Bendixson–Dulac criterion; see e.g. Ch. 3 [3].

2 Reduction to three families

Lemma 3. There is no periodic orbit in $\mathbb{R}^2_{\geq 0}$, for any κ, in each of the following cases: (i) $\operatorname{rank}\Gamma \leq 1$; (ii) $\operatorname{rank}\Gamma = 2$ and $u \notin \mathbb{R}^3_{>0} \cup \mathbb{R}^3_{<0}$; (iii) $u \in \mathbb{R}^3_{<0}$; (iv) $u \in \mathbb{R}^3_{>0}$ and $c_1 \leq 0$; (v) case 9, $u \in \mathbb{R}^3_{>0}$, $c_1 > 0$ and $d_1 = 0$.

Proof. (i) Trajectories lie on lines.

(ii) $u^\perp = \operatorname{im}\Gamma^\top$ contains some $w \geq 0$, $w \neq 0$. If $u_i = 0$ take $w = e_i$; if $u_i > 0 > u_j$ take $w = |u_j|e_i + u_i e_j$. Then $\lambda \cdot (x, y)$ with $\Gamma^\top \lambda = w$ is strictly increasing on $\mathbb{R}^2_{>0}$.

(iii) The unique positive equilibrium is a saddle, of index -1, and it is the only equilibrium inside any periodic orbit in the convex set $\mathbb{R}^2_{>0}$. This contradicts index theory.

(iv) In case 9, $\operatorname{div}(F/(xy)) = (-d_1\kappa_1 x^3 + c_1\kappa_1 x^2 y - c_3\kappa_3 y^2)/(x^2y^2)$. In case 10 it is $(-d_1\kappa_1 x^3 + c_1\kappa_1 x^2 y - c_3\kappa_3 y - d_3\kappa_3 x)/(x^2y^2)$. Both are negative on $\mathbb{R}^2_{>0}$: if $c_1 < 0$ the middle term is negative, and if $c_1 = 0$ then $d_1 \geq 1$. This is Lemma 4 [1].

(v) Here $u > 0$ forces $d_2 \geq 1$, $d_3 = -1$, $c_2 = -1$ and $c_3 = 0$, and then $\operatorname{div}(F/(xy)) = c_1\kappa_1/y > 0$ (cf. §4.2 [1]). □

Lemma 4. If $u \in \mathbb{R}^3_{>0}$, $c_1 > 0$, $d_1 > 0$ and $A > 0$, then BBH's condition holds.

Proof. Case 9. Suppose $c_2 \geq 0$. Then $u_3 > 0$ gives $d_2 \geq 1$, and $u_1 > 0$ gives $c_2, d_3 > 0$. If $c_3 = 0$, then $u_2 < 0$. If $c_3 > 0$, then $d_1 < c_1 d_2/c_2$ and $d_3 > c_3 d_2/c_2$, so $u_2 = c_3 d_1 - c_1 d_3 < 0$. Both contradict $u > 0$, so $c_2 = -1$. Then $u_3 > 0$ reads $d_2/c_2 < d_1/c_1$. If $c_3 > 0$, $A > 0$ reads $\frac{1}{2}(d_3/c_3 + d_1/c_1) < d_2/c_2$. If $c_3 = 0$, $u_2 > 0$ gives $d_3 = -1$.

Case 10. Since $u > 0$, $\sum_i d_i u_i = 0$ and $d \neq 0$, some $d_i < 0$; as $d_1, d_3 \geq 0$, $d_2 = -1$. Then $u_3 > 0$ forces $c_2 = -1$, and $u_1 > 0$ gives $c_3 > d_3 \geq 0$. Conclude as in case 9. □

Lemma 5. If $u \in \mathbb{R}^3_{>0}$, $c_1 > 0$, $d_1 > 0$ and $A \leq 0$, then $c_1, c_3 \geq 1$, $c_2 = -1$, and the network is in exactly one of:

9a $2X \to (2 + c_1)X + d_1 Y$, $X + Y \to 0$, $Y \to c_3 X + (1 + d_3)Y$, with $d_1 > c_1$, $-1 \leq d_3 < c_3$ and $d_1/c_1 + d_3/c_3 \geq 2$;

9b $2X \to (2 + c_1)X + d_1 Y$, $X + Y \to Y$, $Y \to c_3 X$, with $c_3 d_1 \geq c_1$;

10a $2X \to (2+c_1)X + d_1 Y$, $X + Y \to 0$, $0 \to c_3 X + d_3 Y$, with $d_1 > c_1$, $0 \leq d_3 < c_3$ and $d_1/c_1 + d_3/c_3 \geq 2$.

In each, $k := c_3 u_3/(c_1 u_1) \geq 1$, since $A = c_1 u_1 (1 - k)$.

Proof. $A \leq 0 < c_1 u_1$ gives $c_3 \geq 1$, and $c_2 = -1$ as in Lemma 4.

In case 9, $u_1 = -d_3 - c_3 d_2 > 0$ excludes $d_2 \geq 1$. If $d_2 = -1$, we get 9a. If $d_2 = 0$, then $d_3 = -1$ and we get 9b.

In case 10, $d_2 = -1$ and we get 10a. □

Proof (of Theorem 1)**.** In case 10, $u > 0$ and $c_1 > 0$ already force $d_1 > 0$. A network failing BBH's condition is therefore in Lemma 3, or in a family of Lemma 5 by Lemma 4. The families are treated by Theorem 6. □

3 The families: a Liénard argument

Theorem 6. Networks of 9a, 9b and 10a have no periodic orbit in $\mathbb{R}^2_{\geq 0}$, for every κ.

Put $\alpha_1 = c_1 u_1$, $\alpha_2 = u_2$, $\alpha_3 = c_3 u_3$, $\beta_1 = d_1 u_1$, $\beta_2 = -d_2 u_2 \geq 0$ and $\beta_3 = d_3 u_3$. Then $\alpha_2 = \alpha_1 + \alpha_3$, $\beta_2 = \beta_1 + \beta_3$ and $k = \alpha_3/\alpha_1 \geq 1$. By (F3) the system reads

$$\dot{x} = A(x) + B(x)y, \qquad \dot{y} = \rho\big(C(x) + D(x)y\big),$$

with

- case 9: $A = \alpha_1 x^2$, $B = \alpha_3 - \alpha_2 x$, $C = \beta_1 x^2$, $D = \beta_3 - \beta_2 x$;
- case 10: $A = \alpha_1 x^2 + \alpha_3$, $B = -\alpha_2 x$, $C = \beta_1 x^2 + \beta_3$, $D = -\alpha_2 x$ (in 10a $d_2 = -1$, so $\beta_2 = u_2 = \alpha_2$).

One checks:

(1) $BC - AD = u_1 u_2 u_3\, x^2(1-x)$ in case 9, and $u_1 u_2 u_3\, x(1-x^2)$ in case 10.

(2) On $I = (x_0, \infty)$, with $x_0 = \alpha_3/\alpha_2 \in (0,1)$ (case 9) or $x_0 = 0$ (case 10), we have $B < 0$ and $D/B > 0$. In case 9, $D(x_0) = -u_1 u_3$ and $D' = -\beta_2 \le 0$.

(3) The x-nullcline $N = -A/B$ satisfies $N(1) = 1$ and $N'(1) \le 0$.

Periodic orbits lie in $I \times \mathbb{R}$. In case 9, at a minimum $m \le x_0$ of x we would have $\dot{x} \ge \alpha_1 m^2 > 0$.

Lemma 7. Put $f = N' - \rho D/B$, $F = \int_1^x f$, $g = -\rho(BC - AD)/B^2$, $G = \int_1^x g$ and $Y = y - N(x) + F(x)$. Then $\dot{x}/B = Y - F(x)$ and $\dot{Y}/B = -g(x)$. Since $B < 0$ on I, periodic orbits in $I \times \mathbb{R}$ correspond to periodic orbits of $x' = Y - F(x)$, $Y' = -g(x)$. Here $F = N - 1 - \rho\psi$ with $\psi = \int_1^x D/B$ increasing, $f(1) < 0$, $(x-1)g(x) > 0$ on $I\backslash\{1\}$, and $G = \rho\widetilde{G}$ with $\widetilde{G}$ independent of ρ.

Lemma 8. Let $I \ni \bar{x}$ be an open interval, $F \in C^1(I)$ with $F(\bar{x}) = 0$ and $f = F'$, and $g \in C^1(I)$ with $(x - \bar{x})g(x) > 0$ for $x \ne \bar{x}$. Let $G = \int_{\bar{x}}^x g$. Assume $f(\bar{x}) < 0$, and $F(a) > F(b)$ whenever $a < \bar{x} < b$ in I and $G(a) = G(b)$. Then $x' = Y - F(x)$, $Y' = -g(x)$ has no periodic orbit in $I \times \mathbb{R}$.

Proof. Suppose a periodic orbit Γ exists.

Γ *meets* $\{x = \bar{x}\}$ *exactly twice.* Its crossings of the line are transversal, finite in number and at least one; otherwise Y would be monotone. They split Γ into right and left arcs. Each arc is a graph $x = X(Y)$ over an interval containing 0, because Y is strictly monotone on it. Disjoint arcs on the same side are strictly nested, by the intermediate value theorem. If there were two right arcs, the left arc leaving the bottom of the outermost right arc and the left arc arriving at its top would be distinct and not nested, which is impossible. So $\Gamma = R \cup L$, both arcs being graphs over the same interval $[Y_-, Y_+]$, with $Y_- < 0 < Y_+$.

Comparison. Let $Z_{R,L} = G(X_{R,L}(Y))$. Then $Z' = F(X) - Y$, so $\Delta := Z_L - Z_R$ satisfies $\Delta' = F(X_L) - F(X_R)$ and $\Delta(Y_\pm) = 0$. Since $f(\bar{x}) < 0$, we have $F(X_L) > 0 > F(X_R)$ near both ends. Hence $\Delta > 0$ near Y_- and $\Delta < 0$ near Y_+. At the first zero Y^* of Δ after Y_-, we have $\Delta'(Y^*) \le 0$. But there $G(X_L) = G(X_R)$, so the hypothesis gives $\Delta'(Y^*) = F(X_L) - F(X_R) > 0$, a contradiction. □

Since $F(a) - F(b) = [N(a) - N(b)] + \rho[\psi(b) - \psi(a)]$ and the second bracket is positive for $a < b$ (because $D/B > 0$ on I), the hypothesis of Lemma 8 holds for *all* ρ once

$$N(a) \ge N(b) \quad \text{whenever } a < 1 < b \text{ in } I \text{ and } \widetilde{G}(a) = \widetilde{G}(b). \tag{9}$$

Lemma 10. (9) holds in 9a and 9b.

Proof. With $x = (\eta + k)/(1 + k)$, $\eta \in (0, \infty)$, we get $N = (\eta + k)^2/((1 + k)^2\eta)$ and $\widetilde{G} = K\, H_k(\eta)$ with $K > 0$, where

$$H_k(\eta) = \int_1^\eta \frac{(\sigma + k)^2(\sigma - 1)}{\sigma^2}\, d\sigma = \frac{\eta^2}{2} + (2k-1)\eta + (k^2 - 2k)\log\eta + \frac{k^2}{\eta} - \Big(k^2 + 2k - \frac{1}{2}\Big).$$

Let $P(a) = H_k(k^2/a) - H_k(a)$. Then $P'(a) = -Q(a)/a^3$, with $Q(a)/a^2 = 8k(k-1) + (6k-2)s + s^2 > 0$, where $s = (a-k)^2/a > 0$ for $0 < a < 1 \le k$. Also $P(1) = H_k(k^2) \ge 0$. So $P > 0$ on $(0, 1)$.

Hence, if $H_k(\eta_a) = H_k(\eta_b)$ with $\eta_a < 1 < \eta_b$, then $\eta_b < k^2/\eta_a$, i.e. $\eta_a\eta_b < k^2$. Therefore

$$N(a) - N(b) = \frac{(\eta_a - \eta_b)(1 - k^2/(\eta_a\eta_b))}{(1 + k)^2} > 0.$$

□

Lemma 11. (9) holds in 10a.

Proof. Here $g/\rho = (u_1u_3/u_2)\, h'$ with $h = x^2/2 - \log x$. The function $t \mapsto h(1/t) - h(t)$ vanishes at 1 and has derivative $-(1 - t^2)^2/t^3$. So pairs satisfy $ab < 1 \le k$, and $N(a) - N(b) = (a - b)(\alpha_1 - \alpha_3/(ab))/\alpha_2 > 0$. □

Proof (of Theorem 6)**.** Combine (F1), the confinement to $I \times \mathbb{R}$, and Lemmas 7–11. □

Remark 12. In case 10 the computation of Lemma 11 works for every k. For *every* case-10 network with $u > 0$ and $c_1 > 0$, Hopf networks included, $F(a) - F(b) = (b - a)[\rho - (\alpha_1 - \alpha_3/(ab))/\alpha_2]$, and $\operatorname{tr} J < 0 \iff \rho > N'(1)$. Hence there is no periodic orbit whenever $\operatorname{tr} J < 0$. For case-9 Hopf networks we observed the analogous sharpness numerically, but we have not proved it.

In the subfamily of [2] (9a with $d_3 = 0$), $k \ge 1$ is $d_1 \ge 2c_1$. ⌟

4 Computer verification

- `scripts/certify_lienard.py` (sympy, exact; 64 checks) verifies:
  - the normalisation, the equilibrium formulas, $\det J$ and $\operatorname{tr} J$;
  - the Liénard identity for general A, B, C, D;
  - for the symbolic integer data of each family: $u > 0$, (1)–(3), $f(1) < 0$, and $A = \alpha_1 - \alpha_3$ equal to BBH's inequality;
  - the η-substitution, the formula for H_k and for P', the decomposition of Q, the case-10 identities;
  - the Dulac divergences.
- `scripts/classify_networks.py` enumerates all case-9 and case-10 networks whose target coefficients are at most 20 ($1.7 \cdot 10^8$ networks). It confirms that each one falls in exactly one class of Lemmas 3–5 or is a BBH Hopf network, and that BBH's condition is equivalent to $u > 0$, $c_1 > 0$, $d_1 > 0$, $A > 0$.
- `scripts/numerical_sanity.py` (floating point, not part of the proof) finds no cycle

in 128 backward integrations of random family members, finds the cycles of Hopf networks past the bifurcation, and checks the hypothesis of Lemma 8 by quadrature.

The independent verifier's scripts (`verification/`) were written from the source paper before the solver's scripts were read. `classify_independent.py` re-derives the set of non-Hopf networks with $u > 0$ from the trace formula $\operatorname{tr} J = \mu(P/\bar{x} + Q/\bar{y})$ and finds the same classes, with identical counts, in the box of targets at most 20. `verify_symbolic.py` rechecks every identity of this paper on the raw parameters of each family, and proves the key inequality of Lemma 10 through the factorisation $P'(a) = -(a+k)^2((a-1)^2 + k^2 - 1)/a^3$. `verify_numeric.py` (floating point, not a proof) scans the return map on the section $\{x = 1,\ y > 1\}$ of the normalised system for 29 family members, including members with $k = 1$, and for $\rho \in [10^{-4}, 10^{3}]$. It finds no zero of the displacement, while the same detector finds the stable cycle of five Hopf networks just past the bifurcation.

References

[1] Murad Banaji, Balázs Boros and Josef Hofbauer. Oscillations in three-reaction quadratic mass-action systems. *Studies in Applied Mathematics*, **2024**.
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[2] chemistry-open-problems. No limit cycle in the Banaji–Boros–Hofbauer three-reaction example. Problem folder `problems/bbh-three-reaction-no-limit-cycle/` of this repository. Result 2, independently verified 18 October 2026. **2026**. 1, 5

[3] Lawrence Perko. Differential Equations and Dynamical Systems. 3rd ed. Springer, **2001**.
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